

ABAKAS TECHNOLOGIES

Digital Management System Proposal for
Overseas Employment Agencies

A Complete Platform to Manage Workers,
Visas, Documents, Payments, and Reports
Efficiently

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1 Executive Summary

Abakas Technologies presents the Abakas Overseas Employment Management System (AOEMS), a comprehensive digital platform tailored for overseas employment agencies. This system addresses critical operational challenges faced by agencies, transforming manual, inefficient processes into a secure, centralized, and traceable digital ecosystem. Through our observations of real agency operations, we have identified key pain points that hinder productivity and growth.

Our solution empowers agencies to streamline worker registration, visa processing, document management, financial tracking, and partner collaboration, ultimately reducing delays, enhancing transparency, and fostering business expansion. By adopting AOEMS, agencies can expect significant improvements in efficiency, cost savings, and compliance, positioning themselves as leaders in ethical and modern employment practices.

2 Problem Statement

Overseas employment agencies play a vital role in facilitating international labor mobility, particularly between Ethiopia and Arab nations. However, traditional operations rely heavily on manual processes, leading to inefficiencies and operational bottlenecks. Through observation of real agency operations, the following major challenges have been identified:

- ✓ Hundreds of daily phone calls from workers asking about visa status, overwhelming staff and diverting resources from core tasks.
- ✓ Manual paper-based document handling, which is prone to errors, loss, and delays in retrieval.
- ✓ Repeated file sending through messaging apps like WhatsApp, increasing security risks and version control issues.
- ✓ Difficulty tracking worker progress from registration to deployment, resulting in miscommunications and extended timelines.

- ✓ Lack of centralized data across international branches, hindering coordination between Ethiopian agencies and partners in Arab countries.
- ✓ Poor finance history management, including payments, broker commissions, and profits, leading to disputes and inaccurate auditing.
- ✓ Time-consuming report preparation, often requiring manual compilation from disparate sources.
- ✓ Time-consuming CV preparation, which delays partner sharing and selection processes.

These challenges reduce productivity, increase employee stress, and limit business growth. Agencies struggle with scalability, compliance with government regulations (e.g., Musaned and ILMIS), and maintaining worker trust.

3 Proposed Solution: Abakas Overseas Employment Management System (AOEMS)

The Abakas Overseas Employment Management System (AOEMS) is a comprehensive digital platform designed to fully manage overseas employment agency operations—from worker registration to deployment, post-employment monitoring, payments, auditing, and partner collaboration. The system replaces manual, paper-based, WhatsApp-driven, and spreadsheet workflows with a secure, centralized, role-based, and fully traceable digital ecosystem. Built on a modern multi-layer architecture, AOEMS ensures scalability, security, and ease of use.

3.1 How AOEMS Solves the Identified Problems

AOEMS directly addresses each challenge with targeted features:

- 🚦 **Hundreds of daily phone calls from workers asking about visa status:** Workers can independently log in using their passport number and phone number to check their status (e.g., Registered, Visa Approved, Deployed), reducing inquiries and freeing staff time.
- 🚦 **Manual paper-based document handling:** A complete file management system allows secure uploads, downloads, renaming, categorization, previews, and

searches for documents like passports, COCs, contracts, and medical files, eliminating physical paperwork.

- ✚ **Repeated file sending through messaging apps:** The partner notification system automatically alerts partners via in-system notifications when files (e.g., CVs, contracts) are uploaded, providing instant, secure access without external apps.
- ✚ **Difficulty tracking worker progress:** Real-time worker status tracking (e.g., Contract Approved, Medical Completed, Visa Issued) and analytics dashboards provide visibility into progress, with updates triggered automatically at each step.
- ✚ **Lack of centralized data across international branches:** Web-based access for admins, employees, partners, workers, and employers ensures centralized data storage in a relational database and secure cloud storage (Clouinary), enabling seamless cross-border collaboration.
- ✚ **Poor financial visibility:** The payment and financial management module records payments, tracks commissions, calculates profits automatically, and generates audits (monthly/yearly), covering income, expenses, VAT/tax for full transparency.
- ✚ **Time-consuming report preparation:** The reporting system automatically generates customizable reports (e.g., worker details, departure dates, employer info) exportable as PDF or Excel, with analytics including total workers, visa progress, and contract success rates.
- ✚ **Time-consuming CV preparation:** The CV automation system pulls data from the worker database to generate structured CVs in PDF format with one click, including personal info, passport details, skills, experience, and language proficiency.

These solutions collectively enhance productivity, reduce stress, and enable scalable growth

4 System Architecture Overview

4.1 High-Level Architecture

The system follows a modern multi-layer architecture: Client Layer (Frontend): Web-based interface for admins, employees, partners, workers, and employers.

- Application Layer (Backend API): RESTful API with JWT-based authentication and role-based authorization.
- Data Layer (Database & Storage): Relational database for core data and secure file storage in Cloudinary for documents, images, and PDFs.

4.2 Security Layer

- HTTPS encryption.
- HttpOnly cookies.
- Access and refresh tokens for protection against XSS attacks and secure session handling.

5 Authentication & Security Model

AOEMS uses JWT (JSON Web Token) authentication with enhanced security:

- Login via email and password, with forgot/reset password via email.
- Tokens stored in HTTPS-only cookies.
- Access Token: Short-lived for API authorization.
- Refresh Token: Long-lived to generate new access tokens.

This ensures protection against attacks, secure handling, and scalable authentication.

6. Role-Based Access Control (RBAC)

The system is strictly role-based to prevent unauthorized access.

Role	Description
Admin	Full system control
Employee	Operational work

Partner (Saudi)	Read-only assigned access
Worker and Employer	Status check

Table 1

Permission Matrix Each role has predefined permissions. Every API endpoint checks authentication and role authorization, preventing misuse.

7. Public Side Pages

As part of the Abakas Overseas Employment Management System (AOEMS), Abakas Technologies will deliver a professionally designed public-facing website that strengthens the agency's credibility, visibility, and trust with partners, employers, and workers.

These public pages are designed not just for presentation, but to **support business growth, transparency, and brand authority**.

1. Home Page

A high-impact landing page that clearly communicates the agency's services, professionalism, and values.

Includes:

- Hero section with key messaging
- Overview of services
- Agency introduction
- Key achievements and experience
- Call-to-action (Contact / Inquiry)

2. Services Page

A structured presentation of the agency's services, helping partners and clients understand:

- What services are offered
- How the process works

- The agency's areas of specialization

This page improves clarity and reduces inquiry friction.

3. About Us Page

Presents the agency's identity and legitimacy:

- Company background
- Mission and vision
- Ethical employment commitment
- Industry experience

This builds trust and long-term credibility.

4. Experience & Achievements Section

Highlights measurable impact and reliability:

- Years of experience
- Success stories
- Trusted clients and partners
- Workers successfully processed

This section reinforces confidence in the agency's operational capacity.

5. Testimonials Page

Displays feedback and success stories from partners or clients, supporting:

- Social proof
- Reputation building
- Decision-making confidence for new partners

6. Gallery Page

Visual representation of:

- Agency operations

- Training sessions
- Events and facilities

This humanizes the organization and increases transparency.

7. Location Page

Provides clear business presence:

- Office address
- Interactive map
- Working hours

This enhances legitimacy and accessibility.

8. Contact Page

A direct communication channel including:

- Contact form
- Phone numbers
- Email addresses

Designed to convert visitors into inquiries.

9. Social Media Integration

Direct links to official social media platforms to:

- Increase brand visibility
- Strengthen public engagement
- Improve communication reach

Strategic Value to the Agency

By including these public pages, AOEMS enables the agency to:

- Establish a strong professional online presence
- Build trust with international partners

- Increase transparency and ethical credibility
- Support marketing and business development efforts
- Complement the internal management system with a public-facing brand

8. Key Modules and Features

8.1. Worker Registration Module

Employees or admins register workers with validated data fields, including:

- ✓ Personal Information (e.g., name, DOB, photos, education).
- ✓ Passport, COC, and medical info with file uploads.
- ✓ Employment background, skills, language proficiency.
- ✓ Guarantor details.
- ✓ Employer & contract details (Musaned-data).
- ✓ Visa, ILMIS, travel, and ticketing details.
- ✓ Worker status tracking. Workers can check status post-registration.

8.2 CV Automation System

Fully data-driven CV generation:

- ✓ Pulls data automatically from database.
- ✓ Structured sections: Personal info, passport, skills, experience, languages.
- ✓ Output: PDF generated and uploaded on button click.

8.3 Partner Access Workflow:

Partners view and download assigned CVs with read-only access. Notifications ensure smooth sharing.

8.4 Analytics & Dashboard:

Time-based (daily/weekly/monthly) analytics on workers, visas, contracts, and payments, displayed via charts and tables.

8.5 Reporting:

System Generates reports on worker details, status, employers, etc., exportable as PDF/Excel.

8.6 Payment & Financial Management :

Features include payment recording, commission tracking, profit calculation, and audits for income, expenses, profits, VAT/tax.

8.7 Partner Notification:

System Automatic notifications for file uploads to partners.

8.8 Employee Management:

Admins register employees, assign roles, and control access.

8.9 File Management System:

Upload, download, rename, categorize, search, and preview files securely.

9. Security & Data Protection

The system will include:

- Secure login authentication
- Encrypted data storage
- Role-based permissions
- Protection against unauthorized access
- Secure hosting with SSL

Security is a top priority.

10. User Experience & Design

- Clean modern interface
- Mobile-friendly responsive design
- Easily installable PWA (acts like mobile app)
- Fast loading speed (optimized performance)
- Easy navigation

- Minimal training required

Designed for users with limited technical experience.

11. Business Benefits to Overseas Employment Agencies

- Streamlined worker registration and lifecycle management
- Reduced processing delays and operational errors
- Improved transparency and traceability of agency operations
- Centralized and secure document management
- Accurate financial tracking and reporting
- Faster visa and deployment processing
- Reduced reliance on manual, paper-based workflows
- Enhanced compliance with regulatory and ethical standards
- Improved collaboration with partners and employers
- Professional digital presence that builds trust and credibility
- Increased operational efficiency and cost savings
- Strong competitive advantage in the overseas employment market

12. Implementation Approach

1. Requirement Discussion
2. UI/UX Design Approval
3. Database Design
4. System Development
5. Testing & Quality Assurance
6. Deployment
7. Staff Training
8. Ongoing Support

13. SEO & Digital Visibility Strategy

To attract more customers, the website will be developed with:

- SEO-friendly structure
- Optimized page titles
- Clean URLs
- Google indexing
- Mobile optimization
- Fast performance

Benefits:

- ✓ Higher ranking on Google
- ✓ More bookings
- ✓ Strong online presence
- ✓ Long-term marketing advantage

14. Development Timeline

Estimated Total Development Time:

45 Days

Phases:

- ✓ Planning & Design
- ✓ Development
- ✓ Testing
- ✓ Deployment

Timeline may adjust depending on feature additions.

15. Free Training Program

Abakas Technologies will provide **FREE system training** to ensure smooth adoption and confident usage of the Abakas Overseas Employment Management System (AOEMS).

The training will cover the following areas and more:

- Worker registration and profile management
- Visa and process tracking
- Document management and secure record handling
- Financial recording and reporting
- Report generation and performance monitoring
- Partner and employer coordination workflows
- User roles and staff access management

This training ensures that agency staff can effectively use the system from day one, minimizing transition challenges and maximizing operational efficiency.

16. Why Abakas Technologies

- ✓ Strong understanding of overseas employment agency operations
- ✓ Affordable and transparent pricing structure
- ✓ Custom-built system tailored to agency workflows (not generic software)
- ✓ Built using modern, secure, and reliable web technologies
- ✓ Scalable architecture that grows with the agency
- ✓ Long-term support, maintenance, and training
- ✓ We build systems that grow with your agency, streamline operations, and enhance transparency and compliance.

17. Conclusion

The Abakas Overseas Employment Agency Management System is a complete digital transformation platform that ensures operational efficiency, legal accountability, financial clarity, and human dignity for workers. Partner with Abakas Technologies to modernize your agency and drive sustainable growth.